

Abbreviations
~M_1 = TLS12
~M_2 = rs_4
~M_3 = DHE_cipher
~M_4 = group_G
~M_5 = sign(spk(sskS_2),caSKey_1)
~M_6 = group_G
~M_7 = g
~M_8 = exp(g,y_3)
~M_9 = sign((group_G,g,exp(g,y_3),a,rs_4),sskS_2)
~M_10 = ServerHelloDone
~X_1 = ((TLS12,a_3,DHE_cipher,group_G),~M_5,(group_G,g,exp(g,a_4),sign((~M_4,g,exp(g,a_4),~M_12,a_3),~M_14)),ServerHelloDone) = ((TLS12,a_3,DHE_cipher,group_G),sign(spk(sskS_2),caSKey_1),(group_G,g,exp(g,a_4),sign((group_G,g,exp(g,a_4),rc_5,a_3),sskS_2)),ServerHelloDone)
~M_15 = exp(g,x_3)
~M_16 = CCS
~M_17 = PRF(PRF(exp(exp(g,a_4),x_3),master_secret,(rc_5,a_3)),client_finished,Hash(((TLS12,rc_5,DHE_cipher),(TLS12,a_3,DHE_cipher,group_G),(group_G,g,exp(g,a_4),sign((group_G,g,exp(g,a_4),rc_5,a_3),sskS_2)),exp(g,x_3))))
~X_2 = (CCS,PRF(PRF(exp(~M_15,a_4),master_secret,(~M_12,a_3)),server_finished,Hash(((TLS12,~M_12,DHE_cipher),(TLS12,a_3,DHE_cipher,group_G),(group_G,g,exp(g,a_4),sign((~M_4,g,exp(g,a_4),~M_12,a_3),~M_14)),~M_15)))) = (CCS,PRF(PRF(exp(exp(g,a_4),x_3),master_secret,(rc_5,a_3)),server_finished,Hash(((TLS12,rc_5,DHE_cipher),(TLS12,a_3,DHE_cipher,group_G),(group_G,g,exp(g,a_4),sign((group_G,g,exp(g,a_4),rc_5,a_3),sskS_2)),exp(g,x_3))))

A trace has been found.

